



Universidade do Minho

Escola de Engenharia

Departamento de Informática

APRENDIZAGEM AUTOMÁTICA

Machine Learning

Regression

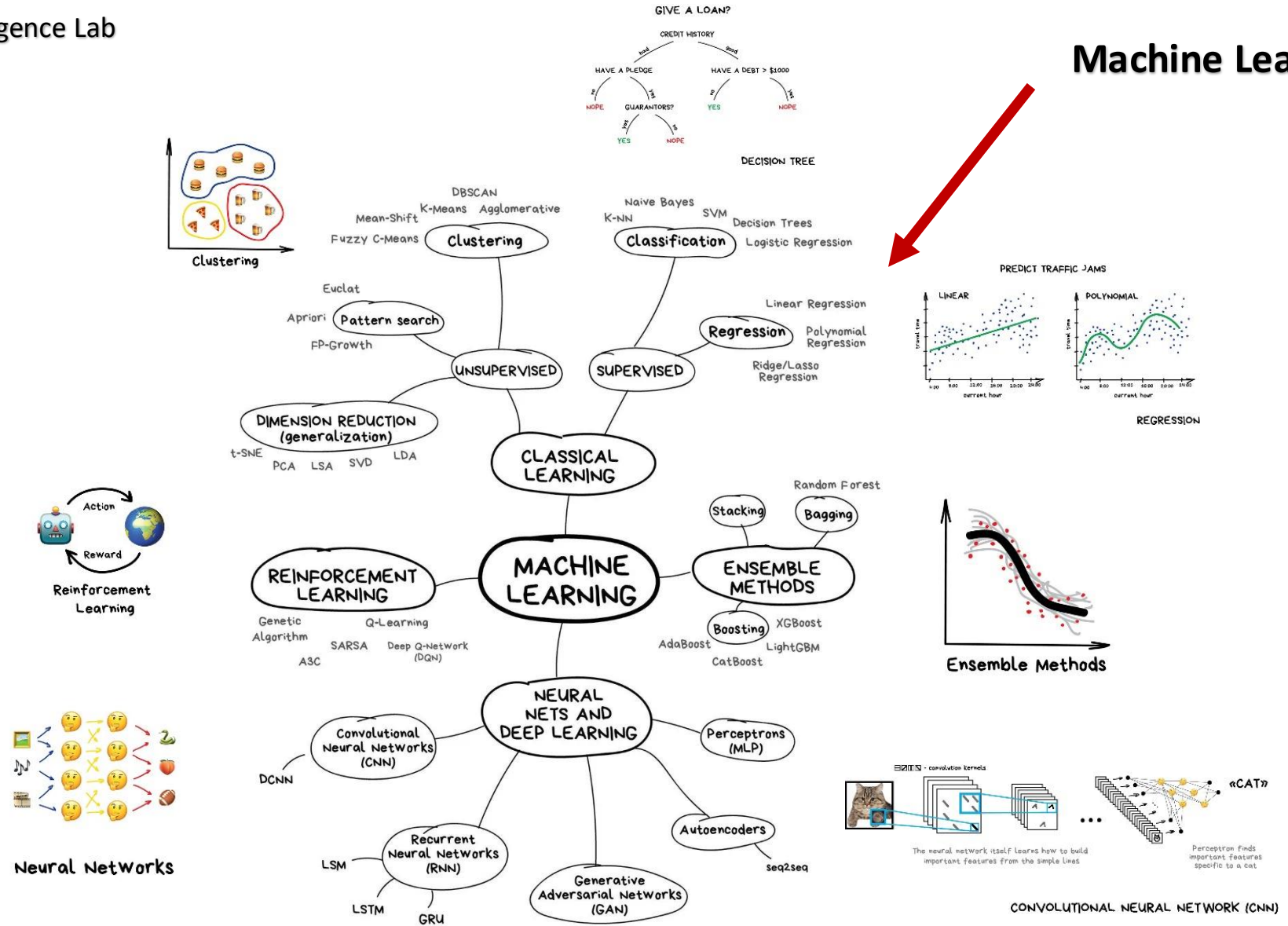
MESTRADO EM ENGENHARIA DE TELECOMUNICAÇÕES E INFORMÁTICA

MESTRADO integrado EM ENGENHARIA DE TELECOMUNICAÇÕES E INFORMÁTICA

Inteligência Artificial para as Telecomunicações

2025/2026

Machine Learning



- **Supervised Learning:**
 - The relationship between variables:
 - Covariance;
 - Correlation.
 - Linear Regression;
 - Multiple regression;
 - Logistic Regression.

The relationship between x and y Covariance

Gives information on the degree to which two variables vary together.

- When $X \uparrow$ and $Y \uparrow$: $\text{cov}(x,y) = \text{pos.}$
 - When $X \downarrow$ and $Y \uparrow$: $\text{cov}(x,y) = \text{neg.}$
 - When no constant relationship: $\text{cov}(x,y) = 0$
- $$\text{cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

The value obtained by covariance is dependent on the size of the data's standard deviations: if large, the value will be greater than if small... even if the relationship between x and y is exactly the same in the large versus small standard deviation datasets.

The relationship between x and y Correlation

- Correlation: is there a relationship between 2 variables?
- Correlation coefficients are indicators of the strength of the linear relationship between two different variables, x and y.
- The correlation coefficient (ρ) is a measure that determines the degree to which the movement of two different variables is associated.

- **Pearson's correlation coefficient** (linear correlation): shows linear correlation between two continuous variables
- **Spearman rank correlation coefficient**: non-parametric alternative to Pearson's correlation coefficient

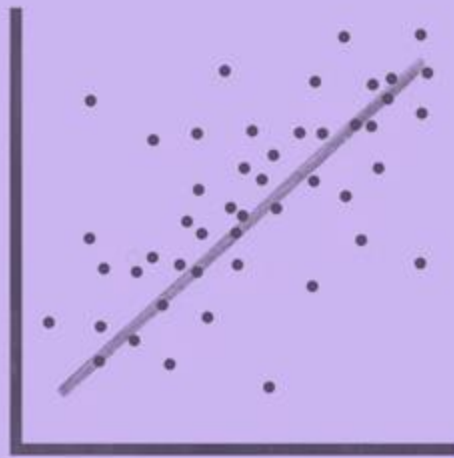
- Measures the relative strength of the *linear* relationship between two variables
 - Unit-less
 - Ranges between -1 and 1
- The closer to -1 , the stronger the negative linear relationship
- The closer to 1 , the stronger the positive linear relationship
- The closer to 0 , the weaker any positive linear relationship



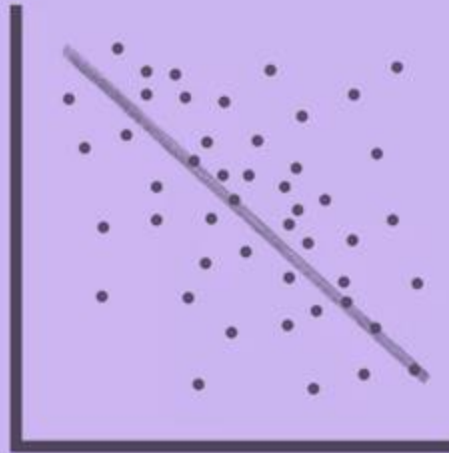
ISLab

Synthetic Intelligence Lab

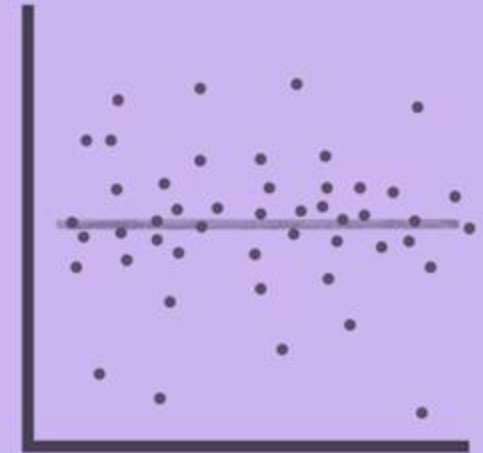
Correlation



Positive Correlation



Negative Correlation



No Correlation

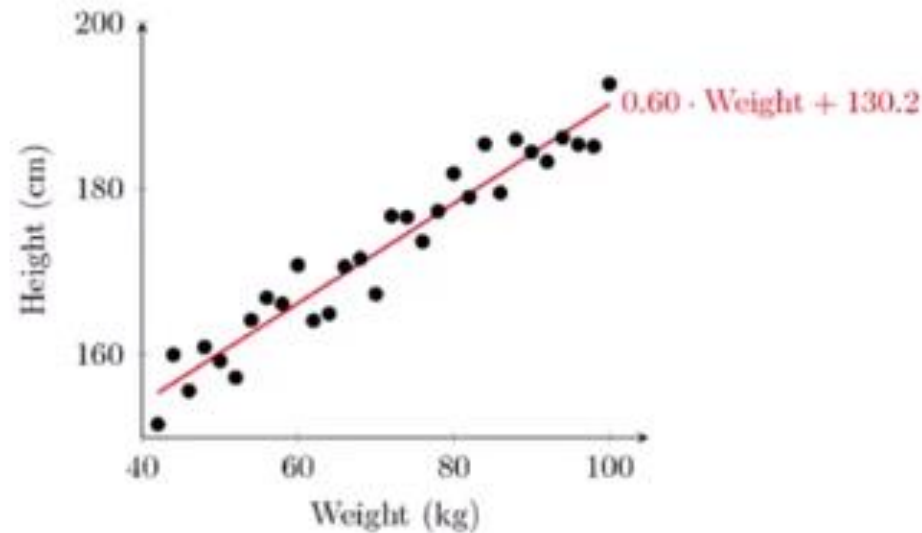
Source image: <https://www.investopedia.com/ask/answers/032515/what-does-it-mean-if-correlation-coefficient-positive-negative-or-zero.asp>

How well a certain independent variable predict dependent variable?

Regression is a statistical procedure that determines the equation for the straight line that best fits a specific set of data.

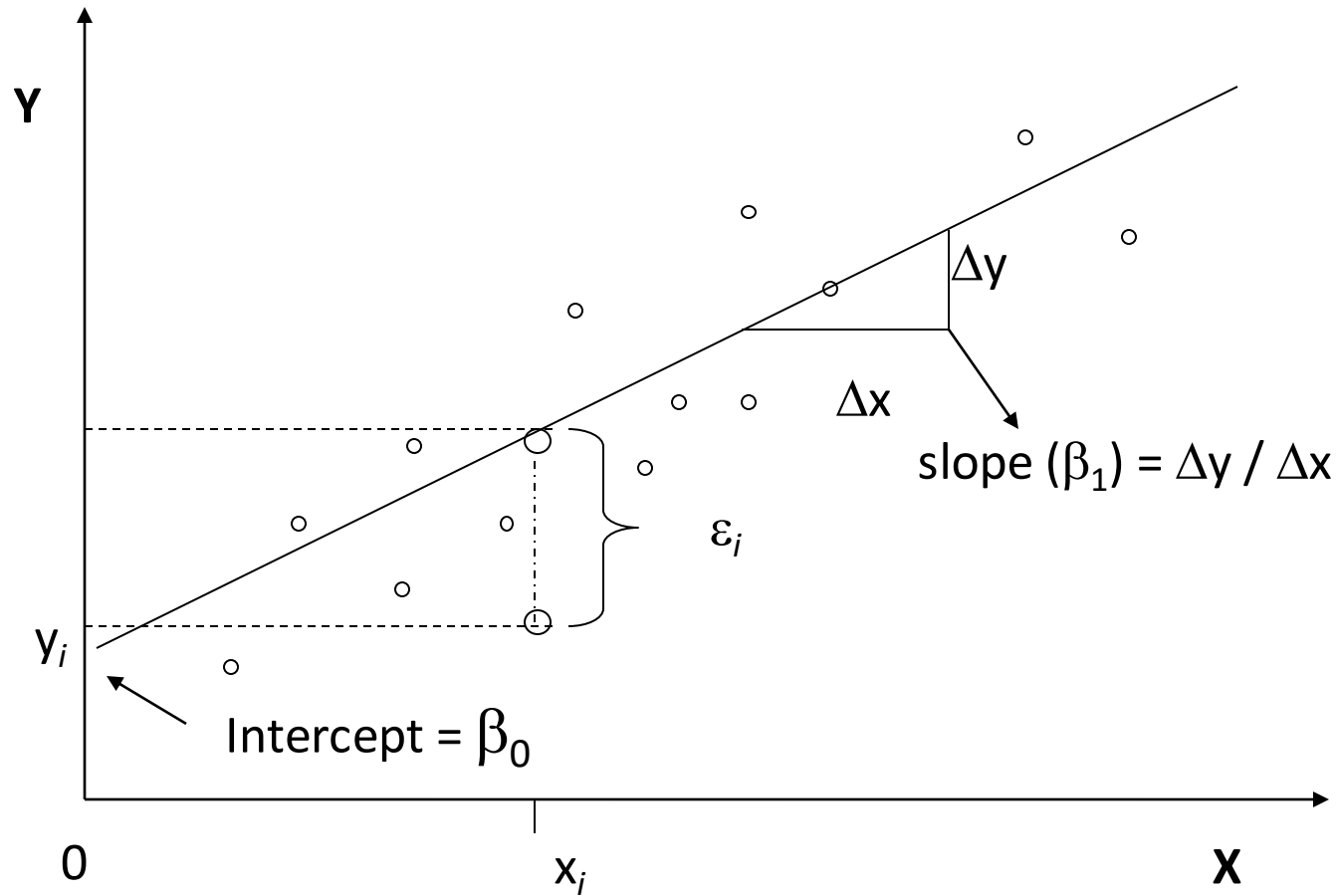
Aims to predict the value of a outcome, Y, based on the value of an predictor variable, X.

- Fit a straight line into a data set of observations;
- Use this line to pred





Linear Regression

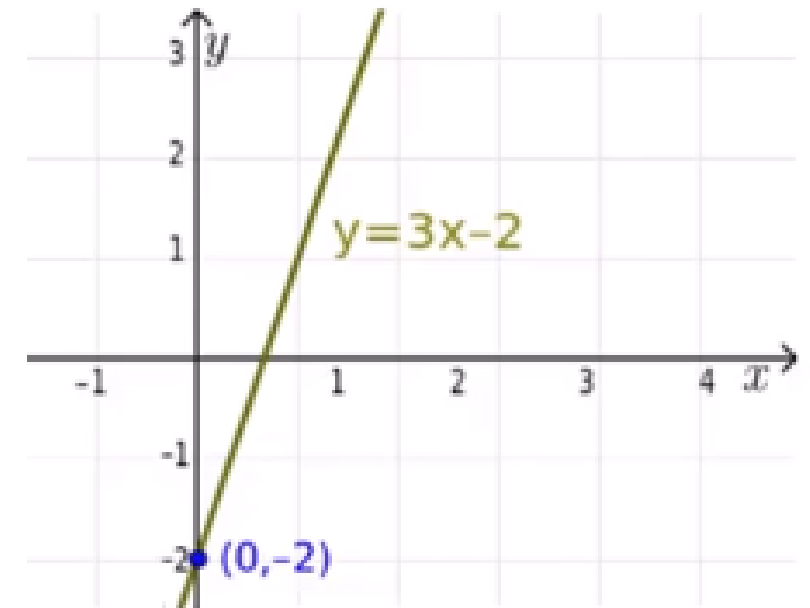


$$\text{Model: } Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

How does it work?

Usually using “least squares” - minimize the squared-error between each point and the line

- Follows the slope-intercept equation of a line: $Y = m \cdot x + b$
 - m – slope: correlation between the two variables times the stand. Dev. In Y , all divided by the standard deviation in X .
 - b – y intercept: Intercept is the mean of Y minus the slope times the mean of X



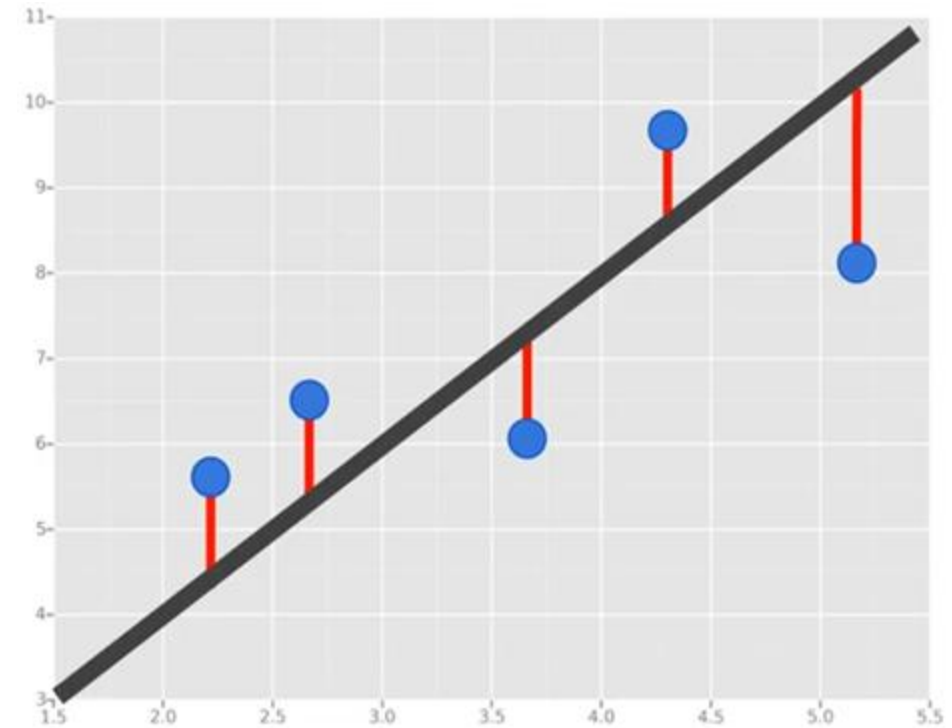
How does it work?

- Least squares minimizes the sum of squared errors
 - $y(i)$: true value
 - $F(x(i), \beta)$: predicted value / fitted line
- Residuals for an observation is the difference between the observation (y-value) and the fitted line

$$r_i = y_i - f(x_i, \beta).$$

The least-squares method finds the optimal parameter values by minimizing the sum, S , of squared residuals:

$$S = \sum_{i=1}^n r_i^2.$$



Measuring error with r-squared

- How do we measure how well our line fits our data?
- R-squared (i.e., coefficient of determination) measures – *“the fraction of the total variation in Y that is captured by the model”*

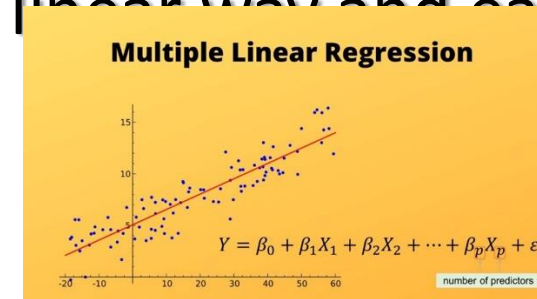
Computing r-squared

- Ranges from [0-1]
- 0 is bad (none of the variance is captured)
- 1 is good (all of the variance is captured)

$$1.0 - \frac{\text{sum of squared errors}}{\text{sum of squared variation from mean}}$$

- Multiple regression is used to determine the effect of a number of independent variables, x_1, x_2, x_3 etc, on a single dependent variable, y
- The different x variables are combined in a linear way and each has its own regression coefficient:

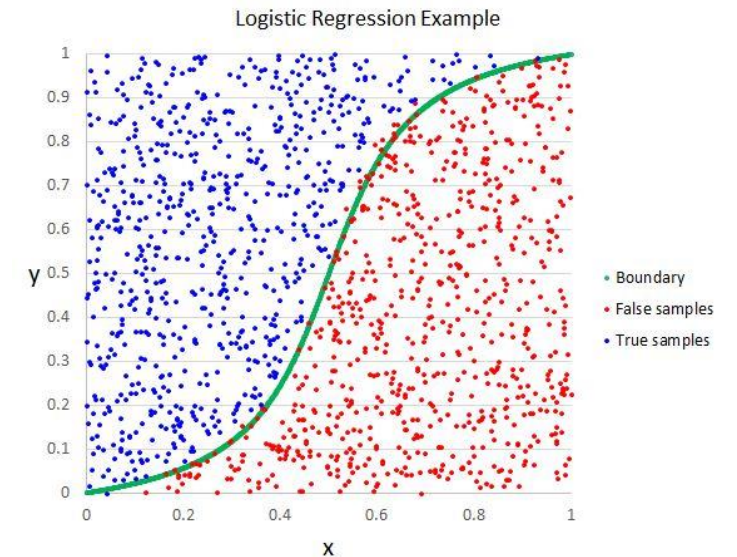
$$y = a_1x_1 + a_2x_2 + \dots + a_nx_n + b + \varepsilon$$



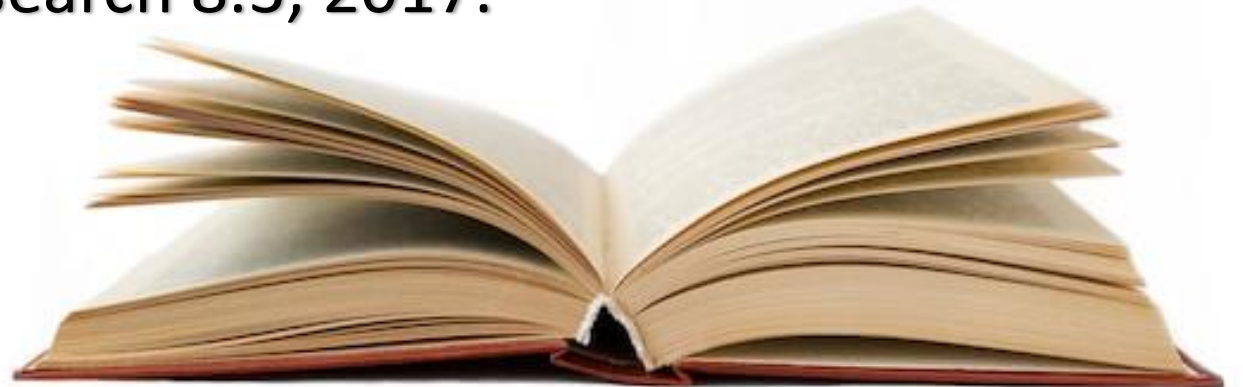
- The a parameters reflect the independent contribution of each independent variable, x , to the value of the dependent variable, y .

Logistic Regression

- The essential difference between linear regression and logistic regression is that the latter is used when the dependent variable is binary in nature.
- In contrast, linear regression is used when the dependent variable is continuous and the nature of the regression line is linear.
- Logistic Regression is a classification technique:
 - Loan (YES/NO)
 - Diagnosis (Are/Sick)
 - Wine (White/Rosé/Red)



- Montgomery, Douglas C., Elizabeth A. Peck, and G. Geoffrey Vining. Introduction to linear regression analysis. John Wiley & Sons, 2021.
- Ranganathan, Priya, C. S. Pramesh, and Rakesh Aggarwal. "Common pitfalls in statistical analysis: logistic regression." Perspectives in clinical research 8.3, 2017.





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